Mission Control — Inspection Brief

Team Agazhvan 2.0 · Agalvaan Mission Control · prepared 17 October 2026
Checked against: 2026 NASA International Lunabotics Championship Guidebook, Version 3 (15 Oct 2026)

What this document is

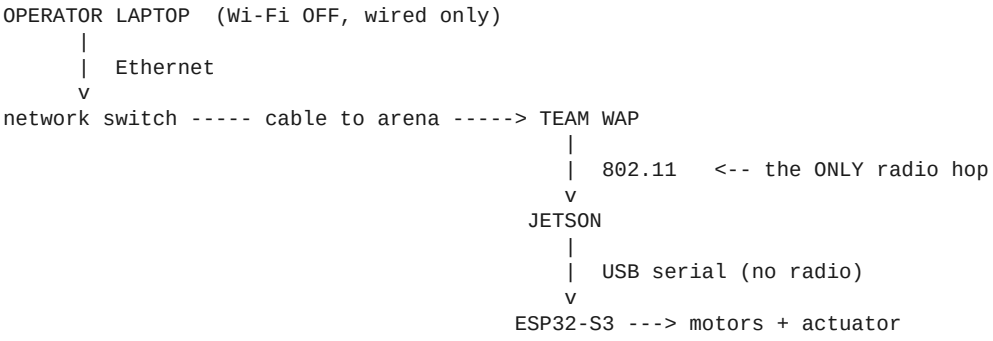
A plain description of how our operator station talks to the rover, what radios are involved, and how we keep the rover safe. It exists because two rules ask us to explain exactly this to judges: **7.2.6** (identify every wireless emitter on the robot) and **5.6.1** (explain how we interact with telemetry during autonomy). Writing it down means the answer is the same every time and does not depend on who is standing at the station.

Items marked **OPEN** are things we have not finished. They are listed on purpose — an inspection brief that hides gaps is not worth reading.

1. The system in one page

Item	Value
Rover computer	NVIDIA Jetson Orin Nano — Ubuntu 22.04, ROS 2 Humble
Motor / actuator controller	ESP32-S3, connected to the Jetson by **USB serial** (not wireless)
Operator station	Laptop running Agalvaan Mission Control (PySide6), Ubuntu 22.04, ROS 2 Humble
Rover address	192.168.2.20
Operator station address	192.168.2.50
ROS 2 domain	ROS_DOMAIN_ID = 17 (identical on both machines)
Competition run length	**25 minutes** (Guidebook v3 §1.3 — reduced from 30)
LiDAR	1 × Livox MID-360 → /livox/lidar
Cameras	D456 DEPTH CAMERA (front) and FRONT CAM (rear)

How the two machines are connected



There is **one** wireless link in the whole system: the team WAP to the Jetson. The operator laptop is wired to the switch and its Wi-Fi radio is switched off (§7.6.6). The ESP32-S3 is wired to the Jetson by USB and emits nothing.

2. Wireless emitters on the robot

Required by §7.2.6. This is the complete list.

Device	Band / use	Notes
Jetson Wi-Fi client	2.4 GHz Ch 1 (20 MHz) or 5 GHz	The only intentional emitter. External antenna fitted (§4.2.4).
ESP32-S3	none — radio unused	Wired USB serial to the Jetson. Wi-Fi/BT not used for any rover function.
Livox MID-360 LiDAR	none (optical)	Ethernet to the Jetson.
Cameras	none	Wired to the Jetson.

No Bluetooth, no Zigbee/802.15.4, no 900 MHz, no power amplifier. Encryption on, SSID broadcast, no channel bonding (20 MHz mask).

3. How we interact with telemetry (§5.6.1)

Mission Control is a **read-only display** for everything except the operator's own deliberate commands. Three things are worth the judges knowing:

	How it works
Telemetry is one-way	Rover state arrives on ROS 2 topics and is drawn on screen. The display cannot write back. Looking at a number sends nothing to the rover.
The display never invents data	If a value is missing or stale, the card reads **-- / NO DATA** rather than holding the last reading. Stale data is never shown as live.
Hands-free means hands-free	During an autonomy attempt no operator touches a laptop or controller. Telemetry keeps updating on its own, which §5.6.1 and §5.6.5 permit.

The rover's reported state is what the screen shows — never what the operator pressed. A button press is a request; the rover's own report is what gets drawn. The interface therefore cannot show a mode the rover is not actually in.

4. Safety chain

Four independent layers. Each stops the rover on its own; none depends on the one above it still working.

Layer	Trigger	Result
1. Physical E-Stop (§4.3)	Operator presses the button on the rover	COTS 40 mm button isolates the batteries. Power logger sits before the kill switch (§4.4.3), so readings survive.
2. Operator stop	E-STOP in Mission Control	Command sent to the rover; the interface latches and shows ENGAGED.
3. Jetson watchdog	No joystick data for **0.5 s**, or no /cmd_vel for 0.5 s in AUTO	Drive and actuator commands are set to stop. Mode changes also stop the rover.
4. ESP32 timeout	No serial command for **500 ms**	Controller stops by itself. This is the layer that protects us if the Jetson, the link, or the USB cable fails.

Commands are re-sent continuously at 10 Hz rather than only when they change, so a single lost serial byte cannot leave the rover driving — the next message corrects it, and silence genuinely means a fault.

Per §4.3.3, releasing the E-Stop restores power only. It does not resume motion; a separate deliberate action is required.

5. Compliance check against Guidebook v3

Rules that affect the operator station and the comms link.

Rule	Requirement	Our status
§1.3 / §6.1.3	Runs are 25 minutes	**COMPLIANT** — Mission clock set to 25:00. Updated from 30 for v3.
§7.6.6	No wireless backchannel from the laptop	**COMPLIANT** — Laptop is wired to the switch; Wi-Fi radio off. On the pre-run checklist.
§7.1.9	20 MHz, no channel bonding on 2.4 GHz	**COMPLIANT** — WAP configured to 20 MHz.
§7.1.4-8	Assigned SSID, broadcast, encrypted	**PARTIAL** — Encryption and broadcast on. SSID must be renamed to the assigned Team_## at check-in.
§7.2.6	Identify all wireless emitters	**COMPLIANT** — Section 2 of this brief.
§5.6.1	Explain telemetry interaction to the MCJ	**COMPLIANT** — Section 3 of this brief.
§4.3.3	E-Stop reset alone must not resume motion	**PARTIAL** — Hardware behaviour correct. The separate arm step in the interface is not built yet.
§4.4.3	Power logger between battery and kill switch	**COMPLIANT** — Wired as required; checked at inspection.
§5.2.4	Arena camera usage is scored	**COMPLIANT** — We run our own front and rear cameras, targeting 0 arena cameras (120 pts/run).
§6.1.2	10-minute setup window	**PARTIAL** — Currently started by hand. Automatic start-up on the rover is not finished.
§6.1.4	Loss of comm ends the run	**COMPLIANT** — Link watchdog at 1 s; the operator sees the link state at all times.
§5.6.4	Autonomy review before the first run	**OPEN** — No autonomy implemented yet, so there is nothing to review. See section 6.
§6.2.6	Each autonomy type scores once per run	**OPEN** — Not applicable until autonomy exists.
§2.5.2	Inspection before every run	**COMPLIANT** — Operational — on the checklist.

6. What is not finished

Listed honestly. None of these is a safety issue; all of them are work still in progress.

Item	Effect	Plan
Autonomy not implemented	The EXCAVATE / DUMP / ONE CYCLE buttons acknowledge but command nothing. They are clearly marked MOCK on screen.	Teleoperation is fully working and is how we intend to run.
Motor firmware timeout not verified	The actuator firmware has a 500 ms timeout. The drive firmware has not been confirmed to have the same.	Confirm before the first run — this is our highest priority item.
Rover does not start itself	Nodes are launched by hand, which uses setup time.	systemd units so the rover is drivable from power-on with no typing.
Topic QoS not tuned	Queue depth 10 can replay a burst of stale commands after a Wi-Fi stall.	Change to best-effort, depth 1, so only the newest command survives.

Generated from the live configuration by `tools/generate_inspection_report.py`. Re-run it after any configuration change so this brief cannot drift from what the software actually does.